End Sem Exam: CHM204: *Electrochemistry, surface chemistry and colloids*
Date (Day) / Time / Place: 6-May-2025 (Tuesday) / 9 AM to 12 Noon / LH5 (LHC)

Answer ALL questions.

Full Marks: 50

1. Provide brief and to-the-point answers to the following: 2 x 10 = 20

i) What are the sources of overpotential in electrolysis of water?

✓ ii) What are the standard reduction potentials for HER and OER reactions in neutral and alkaline media?

✓ iii) What is the difference between Helmholtz and Guoy-Chapman models for the electrical double-layer at an electrode/electrolyte interface?

✓ iv) What are the key differences between the mechanisms Langmuir-Hinshelwood model and the Eley-Rideal model for a surface-catalyzed reaction, say, $2CO + O_2 \rightarrow 2CO_2$

✓ v) Draw a schematic of the potential energy surface for adsorption and dissociative chemisorption when the barrier to chemisorption lies above the zero of potential.

2✓

✓ vi) A solution contains 0.05 M $Al_2(SO_4)_3$. Calculate the ionic strength and mean ionic activity of the solution at 25 ^{0}C.

✓ vii) A soap bubble has a radius of 1.5 mm. Calculate the excess pressure inside it. (Surface tension of soap solution = 0.03 Nm^{-1})

✓ viii) Which ion moves faster in solution: H^+ or K^+? Justify your answer.

✓ ix) Which is more polarizable: a nonpolar molecule with a large size or a small-sized polar molecule? Justify your answer.

✓ x) Explain why real gases deviate from ideal gas behavior at high pressures and low temperatures.

2) Explain the concept of standard electrode potential and its significance in determining the feasibility of a redox reaction. The solubility of barium sulfate ($BaSO_4$) in water at 25°C is 1.0×10^{-5} mol L^{-1} Calculate the solubility product constant (Ksp) for $BaSO_4$ at this temperature.

End Sem Exam: *CHM204: Electrochemistry, surface chemistry and colloids*
Date (Day) / Time / Place: 6-May-2025 (Tuesday) / 9 AM to 12 Noon / LH5 (LHC)

For a pair of electron donor and acceptor, $H_{AB} = 0.03\ cm^{-1}$, $\Delta_r G^0 = -0.182$ eV, and $k_{et} = 30.5\ s^{-1}$ at 298 K. Estimate the value of the reorganization energy.

For a pair of electron donor and acceptor, $k_{et} = 2.02 \times 10^5\ s^{-1}$ when r = 1.11 nm and $k_{et} = 4.51 \times 10^5\ s^{-1}$ when r = 1.23 nm. Assuming that $\Delta_r G^0$ and λ are the same in both experiments, estimate the value of β.

4 + 2 + 4 = 10

3) Calculate the number-average molar mass and mass-average molar mass of a mixture of two polymers, one having M_1= 44 kg mol^{-1} and the other M_2= 88 kg mol^{-1}, with their amounts (numbers of moles) in the ratio 3:2. What average molar mass would be obtained from (a) osmatic pressure (b) light scattering

Calculate the counter length, radius of gyration, and root mean square separation for polypropylene of molar mass 174 kg mol^{-1}. C-C = 153 Å

The volume of oxygen gas at 0°C and 101 kPa adsorbed on the surface of 1.00 g of a sample of silica at 0°C was 0.284 cm^3 at 142.4 Torr and 1.430 cm^3 at 760 Torr. What is the value of V_{mon}?

4 + 3 + 3 = 10

4) Write the reactions for HER and OER in acidic, neutral, and alkaline media.

The transfer coefficient of a certain electrode in contact with M^{3+} and M^{4+} in aqueous solution at 25 C is 0.39. The current density is found to be 55.0 mA cm^{-2} when the overvoltage is 125 mV. What is the overvoltage required for a current density of 75 mA cm^{-2}?

The enthalpy of adsorption of CO on a surface is found to be −120 $kJmol^{-1}$. Estimate the mean lifetime of a CO molecule on the surface at 400 K.

3 + 4 + 4 = 10